

A Modular Systems-Biology Framework for nsPEF-Enabled Extracellular Vesicle Engineering

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Abstract

Nanosecond pulsed electric fields (nsPEF) can stimulate extracellular vesicle (EV) release through coupled membrane electrodynamics, organelle perturbation, Ca^{2+} redistribution, membrane remodeling, repair, vesicle biogenesis, and stress-dependent cargo sorting. However, these processes span nanosecond electrical events, millisecond-to-second Ca^{2+} and organelle signaling, and minute-to-hour EV release and cargo phenotypes, making whole-cell simulation impractical for early model development. Here we define a modular, cell-type-adaptable systems-biology framework for nsPEF-enabled EV engineering. The framework organizes the mechanism into linked layers: pulse delivery and dosimetry; plasma membrane and organelle electrodynamics; ion transport, Ca^{2+} mobilization, ROS, and bioenergetics; Ca^{2+} -dependent remodeling and repair; EV biogenesis and subtype release; cargo sorting, EV composition, and potency; injury, debris, and quality gating; and manufacturing, isolation, and quality-control interfaces. Cell-state and disease-state modifiers act as cross-cutting parameter layers that adapt the shared model to cancer cells, endothelial cells, immune cells, stromal cells, and other EV-producing systems. This structure provides a transparent framework for connecting pulse parameters to EV yield, subtype distribution, cargo state, functional potency, purity, and cell viability.

Keywords

nanosecond pulsed electric fields; electro-exocytosis; extracellular vesicles; EV engineering; systems biology; compartmental modeling; calcium signaling; electroporation; cargo modulation; biomanufacturing

Introduction

Nanosecond pulsed electric fields (nsPEF) are ultrashort, high-intensity electric pulses capable of perturbing plasma membranes and intracellular membrane-bound organelles. Unlike conventional electroporation, which typically uses microsecond-to-millisecond pulses that primarily charge the plasma membrane, nsPEF can access intracellular membranes because the pulse duration is comparable to, or shorter than, relevant membrane-charging times. This physical distinction allows nsPEF to trigger rapid Ca^{2+} mobilization, organelle stress responses, phospholipid redistribution, cytoskeletal remodeling, and regulated or stress-associated vesicle release. nsPEF have emerged as a bioelectric modality in cancer and other biomedical applications, and recent experimental evidence suggests that they can also stimulate robust extracellular vesicle (EV) release, a process referred to here as *electro-exocytosis*¹.

EVs are heterogeneous membrane-bound particles involved in intercellular communication, immune modulation, vascular remodeling, tissue repair, and disease progression. In accordance with current EV nomenclature principles, this manuscript uses the umbrella term EV when vesicle origin has not been experimentally resolved. Mechanistic terms such as exosome, microvesicle, and apoptotic body are used when referring to biogenesis-specific model pathways. This distinction is important for modeling because measured EV populations often reflect mixtures of small EVs, medium/large EVs, apoptotic bodies, damaged organellar fragments, protein aggregates, and

other co-isolated materials. A useful nsPEF-EV model must therefore predict not only total particle release, but also subtype weighting, cargo state, functional potency, purity, and the boundary between controlled EV release and injury-associated debris.

At the cellular level, nsPEF stimulation initiates a cascade that bridges membrane electrodynamics with vesicle biogenesis. Rapid transmembrane-potential changes and nanopore formation enable ionic fluxes, most notably Ca^{2+} influx and redistribution. In parallel, intracellular membranes such as the endoplasmic reticulum (ER), mitochondria, endosomes, and multivesicular bodies (MVBs) may be electrically perturbed, creating additional routes for Ca^{2+} release, ROS generation, trafficking changes, and cargo-sorting modulation. Local Ca^{2+} microdomains activate Ca^{2+} -dependent effectors including calmodulin, annexins,

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calpain, scramblases, phospholipases, synaptotagmin-like sensors, and SNARE-associated fusion machinery. These signals drive phosphatidylserine externalization, cytoskeletal reorganization, membrane curvature, MVB docking and fusion, microvesicle budding, and membrane repair.

From a computational systems-biology perspective, nsPEF-induced EV engineering is best formulated as a modular pathway-level perturbation rather than as a complete whole-cell simulation. In this framework, the electrical pulse is represented as a controllable external input that modifies plasma membrane and organelle permeability, producing time-dependent Ca^{2+} , ROS, ATP, repair, remodeling, trafficking, and injury states. These upstream variables are then mapped to EV release rates, subtype weighting, cargo composition, functional potency, and quality-control outputs. Such a structure preserves mechanistic interpretability while allowing the same core model to be instantiated in cancer cells, immune cells, endothelial cells, mesenchymal stromal cells, or isolated EVs.

The present manuscript therefore focuses on the modeling problem: how to connect an externally defined nsPEF exposure to membrane and organelle perturbation, Ca^{2+} -dependent remodeling, repair, EV biogenesis, cargo modulation, injury-quality gating, and manufacturing-facing EV outputs through a transparent and literature-curatable computational framework.

Methods

We formulated nsPEF-enabled EV engineering as a modular systems-biology framework that links pulse exposure to membrane and organelle perturbation, Ca^{2+} signaling, membrane remodeling, repair, vesicle release, cargo state, injury-quality gating, and manufacturing-facing outputs. Each biological process is represented as a model module with defined inputs, state variables, outputs, and candidate mathematical forms. Paired appendix tables **A1–A16** summarize biological evidence and computational implementation options.

- (i) Define the minimal set of biological and engineering layers required to connect nsPEF exposure to EV yield, subtype distribution, cargo state, potency, purity, and viability.
- (ii) Assign each layer a computational representation appropriate to its dominant timescale, state variables, measurable outputs, and available modeling literature.
- (iii) Separate biological evidence curation from computational model curation for each module.
- (iv) Distinguish cell-based electro-exocytosis from direct post-isolation EV engineering.
- (v) Represent cell type, disease state, activation state, and population heterogeneity as cross-cutting modifiers rather than as a downstream pathway.

Results

The resulting framework is a cell-type-adaptable module map rather than a single parameterized model. Table 1 provides the concise navigation map. Table 2 defines the mathematical modeling strategy and module-interface

logic. The appendix tables then provide the biological and computational evidence-curation structure needed for future model implementation and parameterization.

Discussion

The proposed structure shifts the conceptual endpoint of electro-exocytosis modeling from total EV release alone to nsPEF-enabled EV engineering. In this formulation, EV output is not a single scalar quantity but a multi-dimensional product state that includes particle yield, subtype weighting, cargo composition, potency, purity, and producer-cell viability. This distinction is essential because nsPEF can plausibly increase bona fide EV release, alter cargo sorting, stimulate repair-associated shedding, or generate injury-associated particles depending on pulse dose, cell state, and recovery kinetics. A systems model that does not separate these outputs risks conflating useful EV production with nonspecific stress debris.

A second key feature of the framework is the explicit treatment of organelle electrodynamics. nsPEF effects on the ER, mitochondria, endosomes, and MVBs provide a mechanistic route by which pulse parameters may influence not only Ca^{2+} dynamics but also exosome-biogenesis machinery, cargo sorting, lysosomal routing, and mitochondrial stress. These processes are difficult to resolve experimentally in a single study, but modular modeling allows them to be represented as testable submodels with defined inputs and outputs.

Finally, the framework distinguishes two related but separate applications. In cell-based electro-exocytosis, nsPEF is applied to producer cells to stimulate vesicle release and modify cargo through endogenous signaling and sorting pathways. In direct EV engineering, nsPEF is applied to isolated EVs to transiently permeabilize vesicle membranes for cargo loading or surface modification. These modes share membrane-electroporation principles but require different state variables, size scales, and quality-control outputs. Separating them prevents overextension of a single model while preserving a shared bioelectric modeling backbone.

Acknowledgements

Appendix

Appendix organization and model-curation strategy

The appendix is organized as paired biological and computational tables for each modeling layer. The biological tables document the experimental and mechanistic rationale for including each submodule. The computational tables document candidate mathematical representations, variables, inputs, outputs, and parameterization strategies. Evidence confidence should be updated during literature review using the following working categories: established, supported, hypothesized, or model assumption.

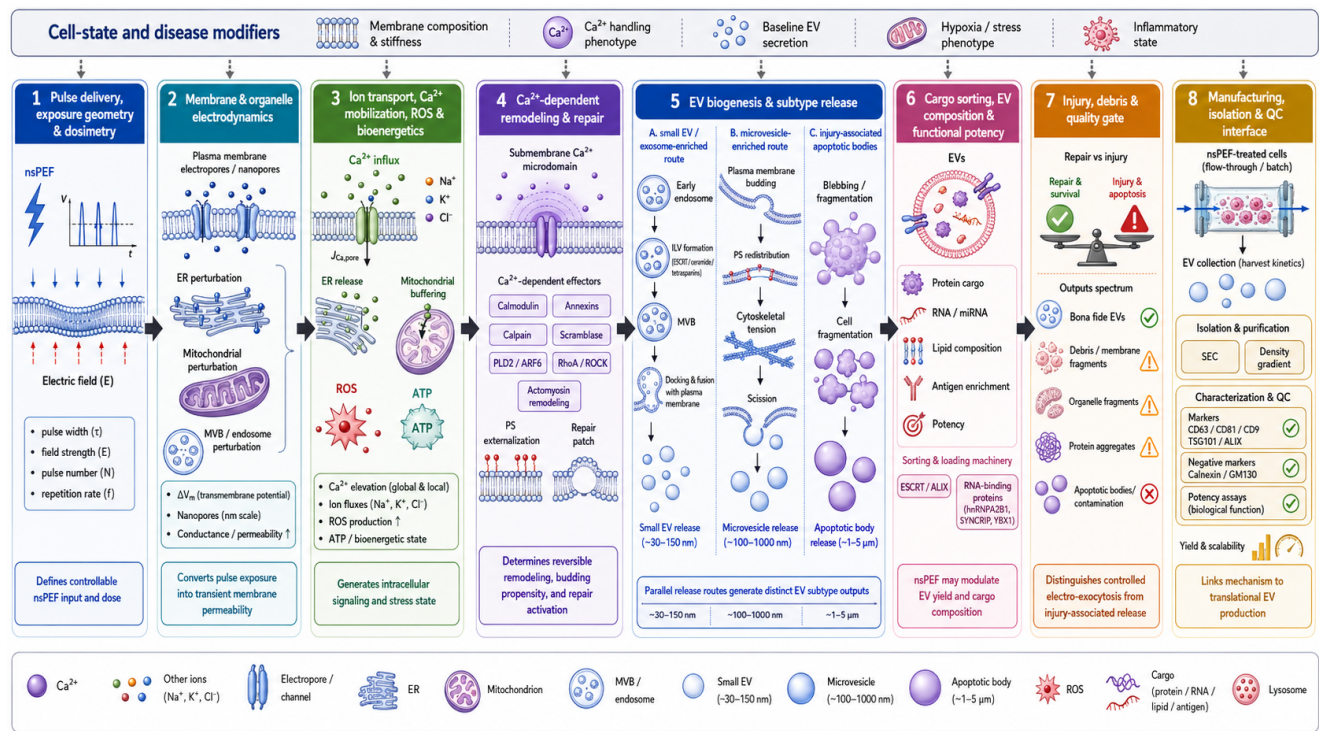


Figure 1. Integrated mechanisms linking nanosecond pulsed electric field stimulation to extracellular vesicle (EV) engineering. Nanosecond pulsed electric fields induce membrane and organelle perturbation, Ca^{2+} redistribution, phosphatidylserine externalization, cytoskeletal remodeling, and repair activation. These processes can promote EV release through endosomal/MVB fusion, plasma membrane budding, and injury-associated vesiculation. In an EV-engineering framework, model outputs include EV yield, size/subtype distribution, cargo composition, functional potency, purity, and cell viability. Abbreviations: EV, extracellular vesicle; ILV, intraluminal vesicle; MVB, multivesicular body; ESCRT, endosomal sorting complex required for transport; VGCC, voltage-gated calcium channel; PLD2, phospholipase D2; ARF6, ADP-ribosylation factor 6; SNARE, soluble N-ethylmaleimide-sensitive factor attachment protein receptor; LAMP1, lysosome-associated membrane protein 1; HIF-1 α , hypoxia-inducible factor 1- α ; IP₃R, inositol 1,4,5-trisphosphate receptor; PS, phosphatidylserine.

Table 1. Concise module map linking nsPEF-enabled EV engineering biology to appendix model-specification tables.

Layer	Module	Model interface/output	Biology table	Computational table
1	Pulse delivery, exposure geometry, and dosimetry	$E(t)$, pulse width, pulse number, repetition rate, energy, temperature	Table A1	Table A2
2	Plasma membrane and organelle electrodynamics	ΔV_m , ΔV_{ER} , ΔV_{mito} , ΔV_{MVB} , $P_i(t)$	Table A3	Table A4
3	Ion transport, Ca^{2+} mobilization, ROS, and bioenergetics	J_{Ca} , C_i , C_{ER} , C_{mito} , ROS, ATP	Table A5	Table A6
4	Ca^{2+} -dependent remodeling and repair	PS exposure, calpain, annexins, actin state, resealing	Table A7	Table A8
5	EV biogenesis and subtype release	R_{sEV} , $R_{m/IEV}$, R_{AB} , subtype weighting	Table A9	Table A10
6	Cargo sorting, EV composition, and potency	protein/RNA/lipid cargo, antigen enrichment, functional potency	Table A11	Table A12
7	Injury, debris, and quality gate	viability, apoptosis, necrosis, debris fraction, contamination markers	Table A13	Table A14
8	Manufacturing, isolation, and QC interface	yield, recovery, purity, batch consistency, scalability	Table A15	Table A16
Cross-cutting	Cell-state and disease modifiers	parameter scalars, phenotype priors, heterogeneity	Tables A1–A15	Tables A2–A16

Abbreviations: nsPEF, nanosecond pulsed electric field; EV, extracellular vesicle; sEV, small EV; m/IEV, medium/large EV; AB, apoptotic body; PS, phosphatidylserine; ROS, reactive oxygen species; QC, quality control.

Pulse Delivery, Exposure Geometry, and Dosimetry

Plasma Membrane and Organelle Electrodynamics

A central goal of the framework is to connect nsPEF exposure to downstream Ca^{2+} -dependent EV release and cargo

modulation without requiring a complete whole-cell reconstruction. The membrane and organelle-electrodynamics layer therefore serves as the upstream bioelectric perturbation module. The applied pulse acts as an external forcing

Table 2. Modeling framework and module-interface logic for the integrated nsPEF-EV engineering model.

Module	Mathematical representation	Inputs	Outputs	Rationale
Pulse delivery and dosimetry	External forcing, exposure geometry, energy/thermal constraints	pulse protocol, chamber geometry, conductivity, cell density	$E(t)$, dose metrics, temperature	Defines controllable engineering input and batch-to-batch exposure.
Membrane and organelle electrodynamics	Circuit model plus pore-density/pore-radius kinetics	$E(t)$, geometry, membrane properties, organelle size	$\Delta V_m, \Delta V_o, G_m, P_i$	Converts nsPEF exposure into plasma-membrane and organelle permeability.
Ion transport, Ca^{2+} , ROS, ATP	Compartmental flux-balance ODEs with optional GHK/Nernst–Planck fluxes	permeability, ion gradients, pumps, channels, stores	$C_i, C_{sub}, C_{ER}, C_{mito}$, ROS, ATP	Links electroporation to signaling, organelle stress, and recovery capacity.
Remodeling and repair	Reaction–diffusion, buffer kinetics, Hill activation, repair ODEs	Ca^{2+} transient, ROS, membrane damage	effector activation, PS exposure, actin state, resealing	Defines reversible remodeling, budding competence, and membrane recovery.
EV release	Pool-balance and Ca^{2+} -dependent rate models	MVB pool, budding sites, Ca^{2+} , PS, cytoskeletal state	$R_{sEV}, R_{m/IEV}, R_{AB}$	Maps molecular states to size/subtype-resolved vesicle release.
Cargo and potency	Cargo-sorting weights, reaction networks, or phenotype-weighted output models	ESCRT, Rab, tetraspanins, stress state, EV subtype	protein/RNA/lipid cargo, potency score	Captures EV engineering beyond yield alone.
Injury and quality gate	Threshold rules, hybrid state transitions, contamination models	residual conductance, Ca^{2+} , ATP, ROS, caspases	viable/stressed/apoptotic fractions, debris burden	Separates useful EV release from injury-associated particles.
Manufacturing and QC	Process-yield balances, recovery models, mixed-effects batch models	harvest time, isolation method, QC markers, scale	yield, recovery, purity, reproducibility	Connects cell-level biology to translational EV production.
Cell-state modifiers	Parameter scaling, priors, mixed-effects or ensemble models	cell type, disease state, activation, stiffness, lipid state	module-specific sensitivity factors	Adapts the shared model to different producer cells and disease contexts.

Note: The table defines modeling form at the module level. Appendix tables provide submodule-specific biological evidence and computational implementation options.

Table A1. Biological evidence table for pulse delivery, exposure geometry, and dosimetry.

Submodule	Biological/process phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Pulse waveform	nsPEF duration, amplitude, number, repetition rate	Parameter-response studies across pulse width, field strength, repetition, and pulse number	Defines controllable input space for EV yield and cargo modulation	Established
Exposure geometry	Electrode/chamber/cuvette/flow geometry	Field uniformity, edge effects, cell density, medium conductivity	Controls reproducibility and batch exposure	Supported
Energy and heating	Joule heating and thermal artifact risk	Temperature traces, energy density, duty-cycle effects	Separates bioelectric effects from heat-induced stress	Supported
Recovery interval	Post-pulse harvest window	EV release kinetics over minutes to hours	Determines when yield, cargo, and viability are measured	Supported
Scale-up mode	Static versus flow-through treatment	Flow exposure, residence time, sterility constraints	Connects mechanism to manufacturing implementation	Hypothesized

Table A2. Computational framework table for pulse delivery, exposure geometry, and dosimetry.

Submodule	Model form	Representative relation	Inputs	Outputs
Pulse input	External forcing function	$E(t) = E_0 u(t; t_p, N, f)$	amplitude, width, number, repetition	$E(t)$
Dose metric	Empirical exposure scalar	$D_E = \sum_k E_k^2 t_{p,k}$	pulse train, conductivity	dose index
Field distribution	Laplace/quasi-static field model	$\nabla \cdot (\sigma \nabla \phi) = 0, E = -\nabla \phi$	chamber geometry, conductivity	local $E(\mathbf{x}, t)$
Thermal constraint	Energy-balance ODE	$\rho c_p \frac{dT}{dt} = \sigma E^2 - h(T - T_0)$	field, medium, cooling	$T(t)$
Flow-through exposure	Residence-time model	$D_E(\mathbf{x}) = \int_{\tau} E(\mathbf{x}(t))^2 dt$	flow rate, chamber field	exposure distribution

function that induces transient plasma-membrane polarization, organelle-membrane polarization, electropore formation, increased conductance, and ion-specific permeability. These outputs provide the primary interface between nsPEF exposure and downstream Ca^{2+} , repair, exocytosis, EV-biogenesis, and cargo-sorting modules.

Ion Transport, Ca^{2+} Mobilization, ROS, and Bioenergetics

The ion-transport and organelle-state module translates nsPEF-induced membrane permeability into intracellular ionic, metabolic, and stress responses. The minimal model treats electropore-mediated Ca^{2+} influx and ER release as source terms in a cytosolic Ca^{2+} balance. More detailed formulations include voltage-gated or leak-channel entry, ER release through IP_3 receptors or ryanodine receptors, SERCA-mediated refilling, PMCA and NCX extrusion, mitochondrial uptake and release, ROS generation, ATP dependence, and osmotic recovery.

$$\begin{aligned} \frac{d[\text{Ca}^{2+}]_i}{dt} = & J_{\text{Ca,pore}} + J_{\text{VGCC}} + J_{\text{ER,rel}} - J_{\text{SERCA}} \\ & - J_{\text{PMCA}} - J_{\text{NCX}} - J_{\text{mito}} - J_{\text{buffer}}. \end{aligned} \quad (\text{A1})$$

Ca^{2+} -Dependent Remodeling and Repair

This module connects global and submembrane Ca^{2+} signals to local membrane remodeling and repair. Although cytosolic Ca^{2+} may be represented as a well-mixed compartment in reduced models, electro-exocytosis is likely governed by local Ca^{2+} domains near pores, damaged membrane regions, budding sites, repair complexes, and fusion machinery. These microdomains activate Ca^{2+} -sensitive effectors that regulate lipid asymmetry, cytoskeletal mechanics, curvature, scission, fusion competence, and resealing.

EV Biogenesis and Subtype Release

The EV-release module converts nsPEF-induced intracellular and membrane responses into measurable vesicle outputs. It integrates endosomal/MVB-derived small-EV release, plasma-membrane budding of medium/large EVs, and apoptotic body or injury-associated vesiculation. The model should retain mechanistic terms such as exosome and microvesicle internally, but report outputs using EV nomenclature compatible with the level of experimental characterization.

Cargo Sorting, EV Composition, and Potency

The cargo module represents the major conceptual expansion from EV-release modeling to EV-engineering modeling. nsPEF may alter EV composition through Ca^{2+} -dependent trafficking, stress-responsive transcription and translation, lipid remodeling, ESCRT/tetraspanin/ceramide pathways, RNA-binding proteins, and injury-state contamination. Therefore, EV output should be represented as a multi-dimensional product state rather than as total particle number alone.

Injury, Debris, and Quality Gate

The injury-quality module defines whether nsPEF exposure produces reversible EV engineering, stress-associated release, apoptosis, necrosis, or contaminating debris. This layer is essential because increased particle number is not necessarily equivalent to increased useful EV production. The quality gate separates bona fide EV output from apoptotic bodies, damaged organellar fragments, protein aggregates, and other co-isolated materials.

Manufacturing, Isolation, and Quality-Control Interface

The manufacturing module links cell-level mechanisms to translational EV production. It represents harvest timing, cell number, culture format, isolation recovery, marker-based QC, purity, batch variability, and flow-through scalability. This module is intentionally placed after biological release and quality gating because manufacturing success depends on both biological production and downstream process recovery.

Cell-State and Disease-State Modifiers as Cross-Cutting Parameters

Cell-state and disease-state modifiers alter how a given cell responds to the same nsPEF exposure. In this framework, they do not define a separate downstream pathway. Instead, they scale or reshape the behavior of all modules. Relevant modifiers include membrane composition, cholesterol and sphingolipid content, cortical stiffness, cytoskeletal prestress, baseline Ca^{2+} handling, mitochondrial state, metabolic capacity, oxidative-stress tone, inflammatory signaling, hypoxia response, baseline EV-biogenesis activity, and cargo-sorting phenotype.

$$\theta^{cell} = S_{cell}\theta^{base}, \quad (\text{A2})$$

$$\theta_i \sim \mathcal{D}(\mu_{cell}, \Sigma_{cell}), \quad (\text{A3})$$

where θ^{cell} represents cell-specific parameter values, S_{cell} represents phenotype-specific scaling factors, and $\mathcal{D}(\mu_{cell}, \Sigma_{cell})$ represents population heterogeneity.

Core Governing Equations for a First Integrated Implementation

A first reduced implementation can be constructed by coupling pulse exposure, membrane permeability, Ca^{2+} balance, remodeling activation, EV-release rates, cargo state, repair, and quality gating. The following equations are placeholders for model assembly and should be refined as literature-supported flux expressions and parameters are identified.

$$C_m \frac{d\Delta V_m}{dt} + G_m(t)\Delta V_m = f(E(t), \theta), \quad (\text{A4})$$

$$C_o \frac{d\Delta V_o}{dt} + G_o(t)\Delta V_o = f_o(E(t), r_o, \theta_o), \quad (\text{A5})$$

$$\frac{dN_p}{dt} = k_{form}(\Delta V) - k_{reseat}N_p, \quad (\text{A6})$$

Table A3. Biological evidence table for plasma membrane and organelle electrodynamics.

Submodule	Biological phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Membrane polarization	Rapid transmembrane-potential rise	Measurements or simulations of ΔV_m under nsPEF exposure	Determines whether local membrane regions reach electroporation range	Established
Plasma-membrane nanopores	Transient nanoscale membrane defects	Conductance, dye uptake, ion permeability, recovery kinetics	Creates pathways for Ca^{2+} and other ion entry	Established
Organelle electroporation	ER, mitochondrial, nuclear, endosomal, and MVB membrane perturbation	Ca^{2+} release, mitochondrial depolarization, intracellular membrane access	Provides route for store release, ROS, and altered MVB/cargo dynamics	Supported
Permeability recovery	Conductance decay and resealing	Patch clamp, uptake assays, dye exclusion	Sets duration and magnitude of post-pulse ion flux	Established
Isolated EV permeabilization	Direct nsPEF effect on EV membranes	Vesicle integrity, cargo loading, aggregation, marker retention	Defines second mode for post-isolation EV engineering	Hypothesized

Table A4. Computational framework table for plasma membrane and organelle electrodynamics.

Submodule	Model form	Representative relation	Inputs	Outputs
Membrane charging	Circuit/capacitive model	$C_m \frac{d\Delta V_m}{dt} + G_m \Delta V_m = f(E(t), \theta)$	$E(t)$, geometry, conductivity	$\Delta V_m(t, \theta)$
Organelle charging	Multi-shell/circuit model	$C_o \frac{d\Delta V_o}{dt} + G_o \Delta V_o = f_o(E(t), r_o, \theta_o)$	organelle size, location, membrane properties	$\Delta V_o(t)$
Pore nucleation	Voltage-dependent kinetic ODE	$\frac{dN_p}{dt} = k_f(\Delta V) - k_r N_p$	ΔV , threshold parameters	$N_p(t)$
Pore-radius evolution	Pore-energy or reduced kinetic model	$\frac{dr_p}{dt} = F(r_p, \Delta V, \gamma, \sigma)$	pore energy, tension	$r_p(t)$
Permeability closure	Conductance/permeability relation	$G_m(t) = G_{m,0} + g_p N_p \pi r_p^2$	pore density, radius	$G_m(t), P_i(t)$
Isolated EV charging	Vesicle-scale circuit model	$C_{EV} \frac{d\Delta V_{EV}}{dt} + G_{EV} \Delta V_{EV} = f(E, r_{EV})$	EV radius, medium conductivity	$P_{EV}(t)$, integrity state

Table A5. Biological evidence table for ion transport, Ca^{2+} mobilization, ROS, and bioenergetics.

Submodule	Biological phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Electropore influx	Ca^{2+} Rapid extracellular Ca^{2+} entry	Ca^{2+} imaging, chelation, Ca^{2+} -free media	Primary trigger for downstream release and repair mechanisms	Established
ER Ca^{2+} release	Intracellular store mobilization	$\text{IP}_3\text{R/RyR}$ involvement, ER store depletion	Amplifies cytosolic Ca^{2+} transients and fusion signals	Established
$\text{Na}^+, \text{K}^+, \text{Cl}^-$ fluxes	Ionic/osmotic perturbation	Membrane-potential recovery, swelling, ionic imbalance	Modulates volume, tension, and repair state	Supported
Mitochondrial Ca^{2+}	Uptake/release and depolarization	Mitochondrial Ca^{2+} , $\Delta\Psi_m$, mPTP evidence	Links Ca^{2+} to metabolic stress and fate decisions	Supported
ROS generation	Oxidative stress after nsPEF	ROS probes, antioxidant rescue, mitochondrial source evidence	Modifies EV secretion, cargo, and injury transition	Supported
ATP/bioenergetics	Energy availability for pumps and trafficking	ATP depletion, pump recovery, metabolic assays	Controls repair, cargo sorting, and EV-release capacity	Hypothesized

$$J_{i,\text{pore}} = P_i(t) \Phi_i([i]_e, [i]_i, V_m), \quad (\text{A7})$$

$$R_{EV} = R_{sEV} + R_{m/LEV} + R_{AB}, \quad (\text{A10})$$

$$R_{sEV} = k_{fus} f_{Ca}([Ca^{2+}]_{sub}) M_d, \quad (\text{A11})$$

$$\frac{d[Ca^{2+}]_i}{dt} = J_{Ca,\text{pore}} + J_{VGCC} + J_{ER,\text{rel}} - J_{SERCA}$$

$$- J_{PMCA} - J_{NCX} - J_{mito} - J_{buffer}, \quad (\text{A8})$$

$$R_{m/LEV} = k_{scission} f_{PS}(PS_{out}) f_{actomyosin} B_{m/LEV}, \quad (\text{A12})$$

$$A_x = \frac{[Ca^{2+}]_{sub}^n}{K_x^n + [Ca^{2+}]_{sub}^n}, \quad (\text{A9})$$

$$Cargo_j = w_{j,sEV} R_{sEV} + w_{j,m/LEV} R_{m/LEV} + w_{j,AB} R_{AB}, \quad (\text{A13})$$

Table A6. Computational framework table for ion transport, Ca^{2+} mobilization, ROS, and bioenergetics.

Submodule	Model form	Representative relation	Inputs	Outputs
Pore-mediated Ca^{2+} flux	GHK/Ohmic flux	$J_{\text{Ca,pore}} = P_{\text{Ca}}(t)\Phi_{\text{Ca}}(C_e, C_i, V_m)$	P_{Ca} , gradients, voltage	Ca^{2+} source term
ER release	Store-release ODE	$J_{\text{ER,rel}} = k_{\text{rel}} O_{\text{IP3R/RyR}}(C_i, C_{\text{ER}}) \frac{C_i^n}{K^n + C_i^n}$	channel state, store content	ER-to-cytosol flux
Pump/exchanger recovery	Saturating fluxes	$J_{\text{PMCA}} = V_{\text{max}} \frac{C_i^n}{K^n + C_i^n}$	Ca^{2+} , Na^+ , ATP, voltage	Ca^{2+} clearance
Mitochondrial buffering	Coupled uptake/release ODE	$\frac{dC_m}{dt} = J_{\text{MCU}} - J_{\text{NCLX}} - J_{\text{mPTP}}$	C_i , $\Delta\Psi_m$, ROS	mitochondrial Ca^{2+} state
ROS dynamics	Source-clearance ODE	$\frac{dR}{dt} = k_{\text{ROS}} f(C_m, \Delta\Psi_m) - k_{\text{clear}} R$	mitochondrial state, antioxidants	ROS state variable
ATP state	Production-consumption ODE	$\frac{dA}{dt} = k_{\text{prod}} f(\Delta\Psi_m) - k_{\text{use}} (J_{\text{pump}} + R_{\text{repair}})$	mitochondria, pumps, repair	ATP availability

Table A7. Biological evidence table for Ca^{2+} -dependent remodeling and repair.

Submodule	Biological phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Ca^{2+} microdomains	Local Ca^{2+} elevation near pores and membranes	High-speed/local Ca^{2+} imaging, buffer sensitivity	Localizes activation to budding, repair, and fusion sites	Supported
Annexin recruitment	Ca^{2+} -dependent membrane binding	Annexin translocation and repair studies	Links Ca^{2+} to membrane repair and curvature stabilization	Supported
Scramblase/PS exposure	Loss of lipid asymmetry	Annexin V staining, scramblase/flippase studies	Promotes MV-like budding and EV surface phenotype	Established
Calpain activation	Ca^{2+} -dependent protease activity	Calpain activation/inhibition, cytoskeletal remodeling	Supports blebbing, scission, and membrane remodeling	Supported
Actomyosin remodeling	Tension, contraction, and scission	RhoA/ROCK, myosin, actin perturbation	Determines budding probability and vesicle detachment	Supported
Lysosomal repair	Ca^{2+} -triggered vesicle fusion	LAMP1 exposure, lysosome fusion	Adds membrane and may release vesicle-like material	Supported

Table A8. Computational framework table for Ca^{2+} -dependent remodeling and repair.

Submodule	Model form	Representative relation	Inputs	Outputs
Ca^{2+} microdomain	Reaction–diffusion or shell ODE	$\frac{\partial C}{\partial t} = D_{\text{Ca}} \nabla^2 C + J_{\text{Ca}} - R_{\text{buffer}} - R_{\text{effector}}$	local flux, diffusion, buffers	$C_{\text{sub}}(t)$
Generic Ca^{2+} sensor	Hill activation	$A_x = \frac{C_{\text{sub}}^n}{K_x^n + C_{\text{sub}}^n}$	local Ca^{2+} , sensitivity	activated fraction
PS exposure	Activation plus lipid-state ODE	$\frac{dPS_{\text{out}}}{dt} = k_{\text{scr}} A_{\text{scr}} (1 - PS_{\text{out}}) - k_{\text{flip}} PS_{\text{out}}$	Ca^{2+} , scramblase, flippase	PS externalization
Calpain/cytoskeleton	Protease activation and substrate loss	$\frac{dF_{\text{actin}}}{dt} = -k_{\text{calp}} A_{\text{calp}} F_{\text{actin}} + k_{\text{poly}}$	Ca^{2+} , calpain state	cortical remodeling
Membrane resealing	Conductance-recovery ODE	$\frac{dG_m}{dt} = -k_{\text{seal}} A_{\text{repair}} G_m$	residual conductance, repair activity	resealed membrane state
Damaged-domain shedding	Damage-removal rate	$R_{\text{shed}} = k_{\text{shed}} D_m f_{\text{Ca}} f_{\text{tension}}$	damaged membrane, Ca^{2+} , tension	repair-associated shedding

Variable and Parameter Glossary Template

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$$\frac{dD}{dt} = k_{inj} G_m + k_{ROS} ROS - k_{repair} A_{repair} D, \quad (\text{A14})$$

$$N_{meas} = N_{EV} + N_{AB} + N_{debris} + N_{agg}, \quad (\text{A15})$$

$$\mathcal{V}_{EV} = \{N_{EV}, R_{EV}, Subtype, Cargo, Potency, Purity, Viability\}. \quad (\text{A16})$$

Table A9. Biological evidence table for EV biogenesis and subtype release.

Submodule	Biological phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Endosome/MVB maturation	Early-to-late endosome and MVB formation	Rab5/Rab7, endosome maturation, ILV evidence	Defines small-EV-producing compartment pool	Established
ESCRT-dependent ILV formation	Cargo sorting and ILV biogenesis	ESCRT, ALIX, TSG101 perturbation	Generates exosome-enriched precursors	Established
ESCRT-independent ILV formation	Ceramide/tetraspanin/lipid-domain sorting	nSMase2, ceramide, CD63/CD81/CD9 evidence	Alternative small-EV biogenesis route	Established
MVB trafficking/docking	Rab-dependent transport to plasma membrane	Rab27A/B, Rab11, Rab35 evidence	Controls release-ready MVB pool	Established
MVB fusion	SNARE-mediated fusion with plasma membrane	SNARE and Ca^{2+} -sensitive fusion evidence	Converts ILVs into secreted small EVs	Supported
Plasma membrane budding	MV-like outward budding	ARF6/RhoA/PS/cytoskeleton studies	Produces medium/large EV precursors	Established
Apoptotic body formation	Blebbing and fragmentation	Caspase/ROCK1/apoptosis evidence	Captures high-dose or injury-associated vesicles	Established

Table A10. Computational framework table for EV biogenesis and subtype release.

Submodule	Model form	Representative relation	Inputs	Outputs
MVB pool	Compartmental pool balance	$\frac{dM}{dt} = k_{mat}E_{endo} - k_{dock}M - k_{deg}M$	endosome pool, maturation	MVB number
ILV loading	Production-rate law	$\frac{dI}{dt} = k_{ILV}f_{ESCRT}M - k_{release}I$	ESCRT/ceramide state, MVB pool	ILV load
MVB docking	Trafficking/docking ODE	$\frac{dM_d}{dt} = k_{dock}f_{Rab}M - k_{fus}M_d$	Rab activity, actin barrier	docked MVB pool
Small-EV release	Ca^{2+} -sensitive fusion rate	$R_{sEV} = k_{fus}f_{Ca}(C_{sub})M_d$	docked MVBs, Ca^{2+} , SNARE	small-EV release rate
Budding-site formation	Budding pool ODE	$\frac{dB_{m/1EV}}{dt} = k_{bud}f_{PS}f_{tension} - k_{scission}B_{m/1EV}$	PS, curvature, tension	budding precursor pool
Medium/large EV release	Scission-rate law	$R_{m/1EV} = k_{scission}f_{actomyosin}B_{m/1EV}$	calpain, ROCK, actomyosin	medium/large EV rate
Apoptotic vesicle release	Injury-state weighted rate	$R_{AB} = k_{AB}F_{apop}$	apoptotic fraction, blebbing rate	apoptotic-body release

Table A11. Biological evidence table for cargo sorting, EV composition, and potency.

Submodule	Biological phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Protein sorting	Ubiquitin/ESCRT/tetraspanin-dependent protein enrichment	ALIX, TSG101, CD63/CD81/CD9, antigen enrichment	Determines therapeutic or immunogenic protein cargo	Established
RNA sorting	RNA-binding protein-mediated miRNA/mRNA loading	hnRNA2B1, SYNCRIP, YBX1, motif-based sorting	Enables nucleic-acid cargo modulation	Supported
Lipid sorting	Ceramide, cholesterol, sphingolipid, PE/PS enrichment	Lipidomics, nSMase2, tetraspanin domains	Controls EV stability, curvature, uptake, immunologic function	Established
Stress-responsive cargo	ROS, hypoxia, inflammatory, apoptotic cargo changes	proteomics/lipidomics/RNA-seq after nsPEF	Distinguishes programmed cargo modulation from debris	Supported
Antigen enrichment	Tumor antigen expression and EV loading	MAGE and other antigen studies	Supports vaccine-oriented EV engineering	Supported
Functional potency	Recipient-cell response	uptake, cargo delivery, immune activation, repair assays	Connects cargo state to biological activity	Hypothesized

Table A12. Computational framework table for cargo sorting, EV composition, and potency.

Submodule	Model form	Representative relation	Inputs	Outputs
Subtype-weighted cargo	Weighted output model	$Cargo_j = w_{j,SEV}R_{SEV} + w_{j,m/LEV}R_{m/LEV} + w_{j,AB}R_{AB}$	subtype rates, cargo weights	cargo abundance
Stress-modulated sorting	Rate law	$\frac{dC_{argo_j}}{dt} = k_{sort,j}f_{stress}f_{Ca}f_{ESCRT} - k_{loss,j}C_{argo_j}$	Ca^{2+} , ROS, ESCRT, stress	cargo state
Antigen enrichment	Transcription/sorting coupling	$\frac{dAg}{dt} = k_{exp}f_{stress} - k_{deg}Ag - k_{sort}Ag$	stress, antigen expression	antigen cargo
Lipid composition	Composition-state ODE	$\frac{dL_j}{dt} = k_{syn,j}f_{stress} - k_{sort,j}L_j$	lipid metabolism, EV subtype	lipid cargo class
Potency mapping	Empirical or mechanistic response model	$Potency = F(Cargo, Subtype, Dose)$	cargo vector, recipient assay	functional potency
Direct EV loading	Permeability-diffusion model	$\frac{dC_{load}}{dt} = P_{EV}(t)A_{EV}(C_{ext} - C_{EV}) - k_{leak}C_{EV}$	EV permeability, cargo gradient	loaded cargo

Table A13. Biological evidence table for injury, debris, and quality gating.

Submodule	Biological phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Viability preservation	Reversible response versus cell death	viability, membrane recovery, clonogenic/metabolic assays	Defines acceptable EV-production window	Established
Mitochondrial injury	Depolarization, mPTP, cytochrome c	$\Delta\Psi_m$, cytochrome c, ROS	Separates stress signaling from organellar contamination	Supported
Apoptosis	Caspase activation and apoptotic blebbing	caspase, annexin V/PI, ROCK1 blebbing	Determines apoptotic-body contribution	Established
Necrosis/lysis	Irreversible membrane failure	LDH release, PI uptake, debris microscopy	Indicates unacceptable product contamination	Established
Non-EV contaminants	Protein aggregates, organelle fragments, apoptotic bodies	calnexin, cytochrome c, albumin/lipoprotein markers	Defines purity and interpretability of EV readouts	Established
Quality-control marker panel	Positive and negative EV markers	CD63/CD81/CD9, TSG101/ALIX/syntenin, negative markers	Connects model to MISEV-style QC	Established

Table A14. Computational framework table for injury, debris, and quality gating.

Submodule	Model form	Representative relation	Inputs	Outputs
Damage accumulation	Damage-state ODE	$\frac{dD}{dt} = k_{inj}G_m + k_{ROS}ROS - k_{repair}A_{repair}D$	conductance, ROS, repair	damage burden
Fate transition	Hybrid state rule	$F_{apop} = H(D - D^*)$ or $\frac{dF_{apop}}{dt} = k_{apop}H(D - D^*)$	damage, mitochondria	ATP, viable/stressed/apoptotic fractions
Necrotic transition	Threshold rule	$F_{nec} = H(D - D_{nec}^*)H(A_{ATP}^* - ATP)$	damage, ATP, membrane failure	ATP, necrotic fraction
Measured particles	Mixture model	$N_{meas} = N_{EV} + N_{AB} + N_{debris} + N_{agg}$	release rates, injury state	measured particle count
Purity score	Ratio or classifier	$Q_{EV} = \frac{N_{bona fide EV}}{N_{meas}}$	marker panel, debris state	EV quality score
Acceptable window	Constrained optimization	maximize Y_{EV} subject to $Q_{EV} > Q^*$ and $Viability > V^*$	dose, cell state, harvest	operating region

Table A15. Biological/process evidence table for manufacturing, isolation, and quality-control interface.

Submodule	Process phenomenon	Evidence to curate	Relevance to EV engineering	Confidence
Harvest kinetics	Time-dependent EV accumulation	minute-to-hour release curves after nsPEF	Determines optimal collection window	Supported
Cell-normalized yield	EV output per cell or protein	NTA/flow/ELISA normalized to cell number	Defines productivity	Established
Isolation recovery	SEC, ultracentrifugation, density gradient, affinity capture	recovery and purity comparisons	Converts released EVs into isolated product	Established
QC characterization	Size, morphology, markers, contaminants	NTA, cryo-EM/TEM, western blot/flow, proteomics	Determines whether product meets EV criteria	Established
Batch variability	Donor, passage, culture, and pulse variability	repeated-batch studies, mixed-effects data	Determines reproducibility and manufacturability	Supported
Scalable flow processing	Closed-system pulse chamber and inline monitoring	field uniformity, sterility, temperature, flow control	Enables translational manufacturing	Hypothesized

Table A16. Computational framework table for manufacturing, isolation, and quality-control interface.

Submodule	Model form	Representative relation	Inputs	Outputs
Cell-normalized yield	Productivity metric	$Y_{EV} = \frac{N_{EV}}{N_{cells}}$	EV count, cell number	EV yield per cell
Harvest accumulation	Time-integral model	$N_{EV}(t_h) = \int_0^{t_h} R_{EV}(t) dt$	release rate, harvest time	collected EVs
Isolation recovery	Process-yield balance	$N_{iso} = \eta_{iso} N_{released}$	isolation method, recovery	isolated EV count
Purity metric	Product-quality ratio	$Purity = \frac{N_{EV}}{Protein_{total}}$ or marker-based score	particle/protein/marker data	purity estimate
Batch model	Mixed-effects model	$Y_b = \mu + \alpha_{cell} + \beta_{dose} + u_b + \epsilon$	cell state, dose, batch	reproducibility
Optimization objective	Constrained process optimization	maximize $Potency \times Y_{iso}$ subject to purity and viability constraints	dose, harvest, isolation	optimal protocol

Table A17. Variable and parameter glossary for the nsPEF-EV engineering model.

Symbol	Definition	Units	Module	Source/estimation strategy
$E(t)$	Applied electric-field waveform	V/m	Dosimetry	Pulse protocol/field model
D_E	Electrical dose index	variable	Dosimetry	Derived from pulse protocol
ΔV_m	Plasma-membrane transmembrane potential	V	Electrodynamics	Circuit/electroporation model
ΔV_o	Organelle membrane potential perturbation	V	Organelle electrodynamics	Multi-shell/circuit model
N_p	Pore density or pore number	pores/ μm^2 or count	Electroporation	Fit or literature prior
r_p	Representative pore radius	nm	Electroporation	Literature prior or fit
G_m	Effective membrane conductance	S/ m^2	Membrane/repair	Fit from permeability recovery
P_i	Ion-specific membrane permeability	m/s	Ion transport	Estimated from flux data
$J_{Ca,pore}$	Electropore-mediated Ca^{2+} flux	concentration/time	Ion transport	Derived from permeability relation
$[Ca^{2+}]_i$	Cytosolic Ca^{2+} concentration	M	Ca^{2+} signaling	Ca^{2+} imaging/model state
$[Ca^{2+}]_{sub}$	Submembrane Ca^{2+} concentration	M	Microdomain	Reaction-diffusion/shell model
ROS	Reactive oxygen species state	variable	Stress/bioenergetics	ROS imaging/probes
ATP	Energy availability state	variable	Bioenergetics	ATP assays/model state
PS_{out}	Externalized phosphatidylserine fraction	dimensionless	Remodeling	Annexin V or model state
M_d	Docked MVB pool	count or concentration	EV release	Fit/estimated latent state
$B_{m/lEV}$	Medium/large EV budding precursor pool	count or concentration	EV release	Fit/estimated latent state
R_{sEV}	Small-EV release rate	particles/time	EV output	NTA/flow/marker-based fit
$R_{m/lEV}$	Medium/large EV release rate	particles/time	EV output	NTA/flow/marker-based fit
R_{AB}	Apoptotic body release rate	particles/time	Injury output	Apoptosis-weighted output
$Cargo_j$	Abundance of cargo class j	variable	Cargo	Proteomics/RNA/lipidomics
$Potency$	Functional activity metric	assay-specific	Potency	Recipient-cell assay
D	Integrated damage burden	dimensionless	Quality gate	Hybrid state variable
Q_{EV}	EV quality/purity score	dimensionless	QC	Marker/purity model
η_{iso}	Isolation recovery efficiency	dimensionless	Manufacturing	Process validation